CS 101: Lecture Notes · Week 4

Cryptographic Document Fingerprinting and HMAC

1. Threat Model & Attacker Capabilities

When confidential materials are distributed to multiple recipients, malicious actors may leak documents or attempt to frame innocent peers. A robust fingerprinting scheme must bind recipient identities with unforgeable authenticity.

Cryptographic Document Fingerprinting (Page 2)

2. HMAC-SHA256 vs Plain Hashing

Plain hashing H(doc_uid || student_uid || nonce) is insecure against framing attacks because any adversary who discovers the nonce can compute candidate hashes. HMAC-SHA256 requires knowledge of the server master key K_master, ensuring mathematical unforgeability under chosen-message attacks.

Cryptographic Document Fingerprinting (Page 3)

3. Dual Carrier Redundancy

To survive common PDF processing workflows (such as PyMuPDF resaves, PDF optimization, and metadata stripping), SecurePDF embeds fingerprints across two independent layers:

· Layer 1: PDF Document Catalog Dictionary (/SPDFPayload)
· Layer 2: Micro-scale Invisible Annotations across content pages

If an adversary strips either layer, the surviving carrier permits full attribution.